



Amazon Bedrock
AgentCore

Amazon Bedrock AgentCore

Build & Deploy AI Agents on AWS [HANDS-ON]

Amazon Bedrock AgentCore

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***Section on :
Course Introduction***





*Course Outline & Few Important Tips –
Promise to complete in few mins 😊*

Course Outline and Important Tips

1. The focus of this course is to help you learn how to **Build, Deploy & Monitor Agentic AI apps on Amazon Bedrock AgentCore**
2. We will be building a **Personal Vacation Planner app using CrewAI** before deploying to Amazon Bedrock AgentCore
3. Pre-Requisites – **Need basic AWS and Python knowledge.**

Generative AI on AWS - Amazon Bedrock, RAG & Langchain[2025]

<https://www.udemy.com/course/amazon-bedrock-aws-generative-ai-beginner-to-advanced/?referralCode=A02153572B8864F928E7>

Build Production Ready AI Agents on AWS - Bedrock, CrewAI & MCP

<https://www.udemy.com/course/aws-ai-agents-complete-course-hands-on/?referralCode=80740CF274CB92574FF8>

Course Outline and Important Tips

1. Course Content Download and Course Delivery Speed
2. This is a new course, incase of any discrepancies or issue with videos/missing content, please mail me at trisalrahul@gmail.com
3. Connect - www.linkedin.com/in/rahul-trisal or Instagram - [Rahul Trisal](#)
4. Course Cost – **Warning** : Less than \$1-2 if you clean up after Hands-On. **High charges for AgentCore Gateway.**
5. Share your success – Tag me on LinkedIn for wider reach
6. Course Rating and Feedback



Course Rating and Feedback – Super Important !!!

Udemy | AWS AI Agents - Complete Course [HANDS-ON]

★ Leave a rating Your progress

Course content AI Assistant Beta

Section 4: Use Case 1 - Hotel Booking Agent
1 / 17 | 2hr 17min

- ☐ 17. Business Use Case - Hotel Booking Agent
2min Resources
- ☐ 18. Use Case Solution Architecture
6min
- ☒ 19. Agent Creation
9min Resources
- ☐ 20. Agent Integration with Knowledge Bases for Room Information
13min Resources
- ☐ 21. Hotel Room Availability - Arch and DynamoDB creation

AI Agents – Key Characteristics

The diagram illustrates the key characteristics of AI Agents. At the center is the 'Agent (Powered by LLM)'. To its left, a 'Tools' box lists various capabilities: 'Tool to access DB', 'RAG - SharePoint', 'Code Interpreter', 'Weather API', and 'Travel Booking'. An example 'AWS Lambda' is noted. The 'Tools' box is connected to the 'Agent'. Above the 'Agent' are 'Short Term Memory' and 'Long term Memory' boxes. To the right of the 'Agent' is a 'Planning' box, which connects to an 'Action' box at the bottom. The 'Planning' box also connects to a 'Chain of Thought' box, which further connects to a 'Tree of Thought' box, and finally to a 'ReAct' box. A large play button is overlaid on the 'Agent' box.

How would you rate this course?

Select Rating



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Course Rating and Feedback – Super Important !!!

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AI Agents – Key Characteristics

AI Agents

Example : AWS Lambda

Tools to access DB, RAG - SharePoint, Code Interpreter, Weather API, Travel Booking

Tools

Agent (Powered by LLM)

Short Term Memory, Long term Memory

Planning

Chain of Thought, Tree of Thought, ReAct

Action

Guardrails – Toxicity, Hallucination and PII Redaction

Agent Communication

0:00 / 0:00

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Section on :
***Bedrock AgentCore - Basic Concepts &
Architecture***

What is Amazon Bedrock AgentCore & the Problem it is trying to solve ?

Enterprise Challenge with Agentic AI

- How to transform Agentic AI apps from “cool demos” into “production-ready business solutions”

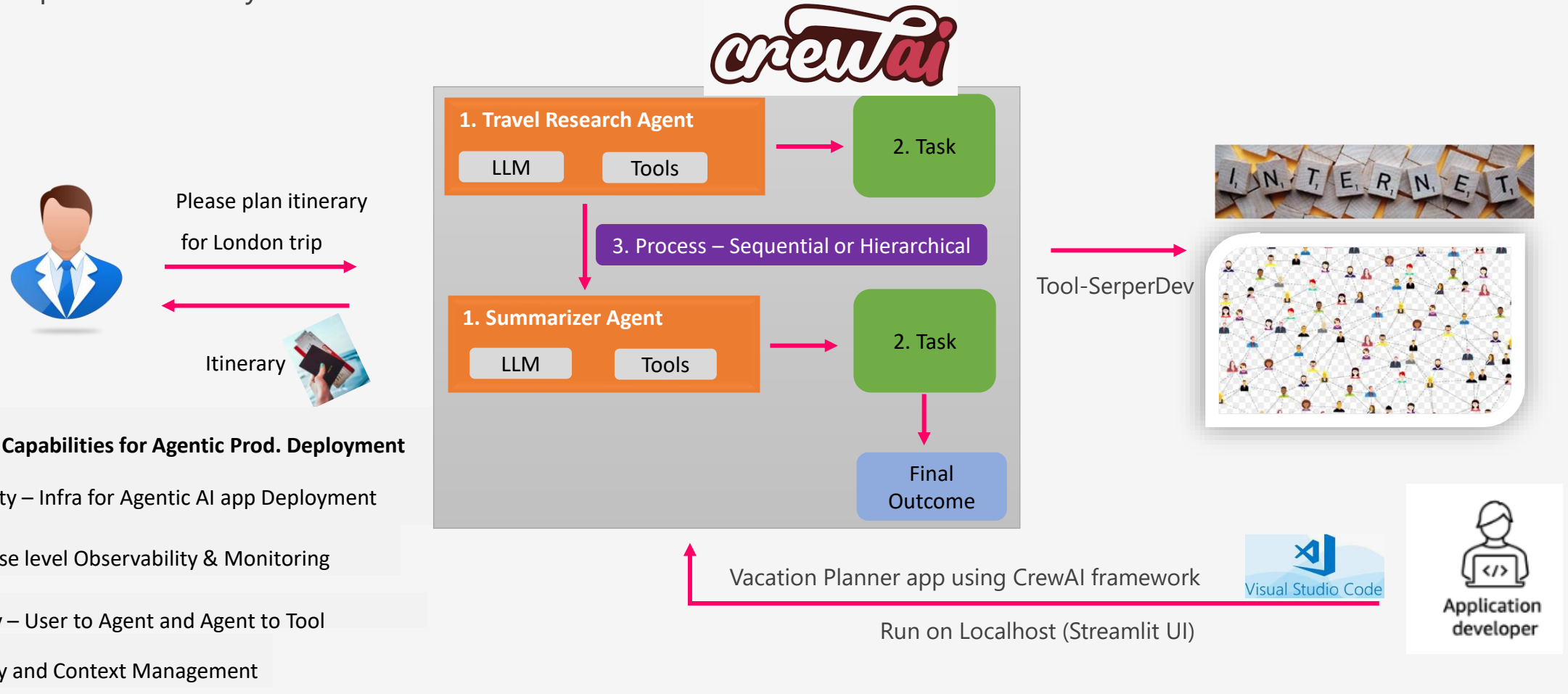
Amazon Bedrock AgentCore Service

- Provides a **set of capabilities** to transition your agentic AI applications from experimental proof of concept to production-ready systems.

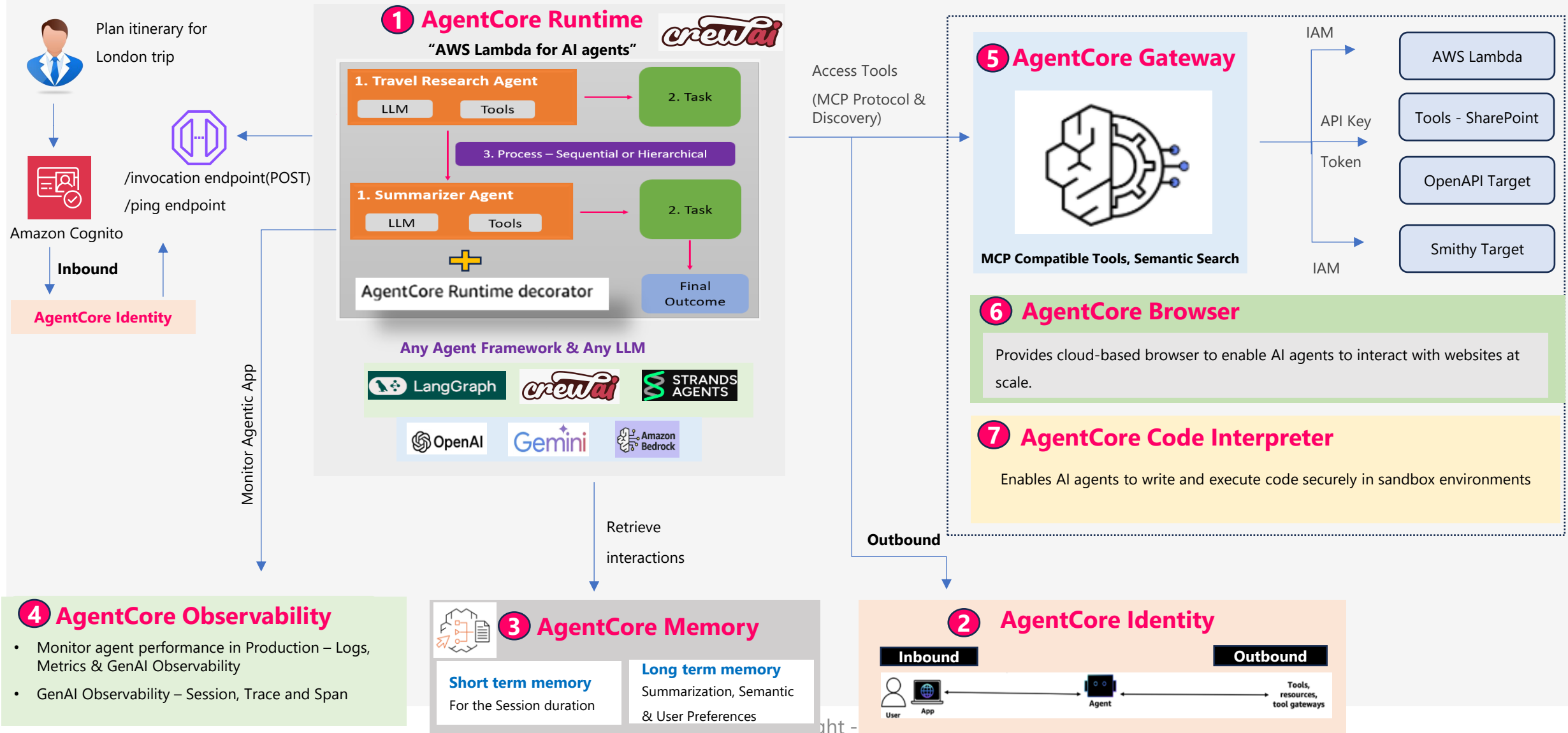


Bedrock AgentCore – Business Scenario : PoC to Production

Business Use Case : Dream Vacation Planner travel agency wants to build and test (PoC) an Agentic AI app which allows its users to plan an itinerary for their dream destination on their website.



Bedrock AgentCore Capabilities – Overview





Section on :

Build a Personal Vacation Planner

Agentic AI App

What is CrewAI ?

[Home](#) [Enterprise](#) [Open Source](#) [Ecosystem](#) [Case Studies](#) [Templates](#) [Use Cases](#) [Blog](#)

[Log in](#)

[Start Enterprise Trial](#)

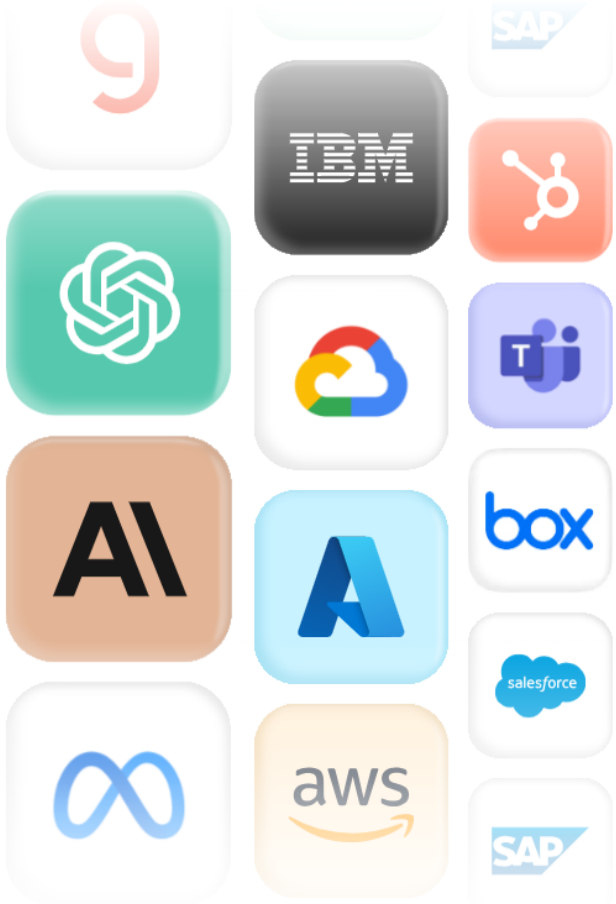
The Leading Multi-Agent Platform

Streamline workflows across industries with powerful AI agents. Build and deploy automated workflows using any LLM and cloud platform.

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500,000,000+
Multi-Agent Crews run using CrewAI



Key Agentic AI Platform

Open Source

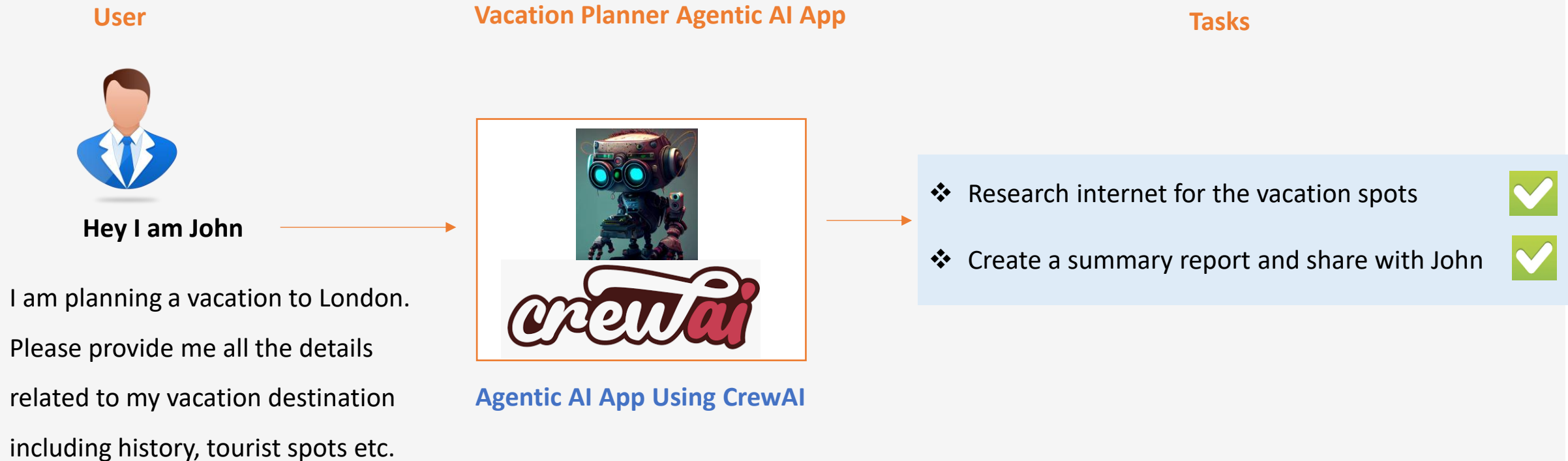
- Crew AI
- LangChain/LangGraph

Open Source - AWS

- Strands

CrewAI is a lean, lightning-fast Python framework built entirely from scratch—completely independent of LangChain or other agent frameworks.

Business Use Case – Agentic AI App using CrewAI



Agentic App using CrewAI – Architecture



4. Crew



Please plan itinerary for
London trip

1. Travel Research Agent

LLM

Tool – SerperDev

Search
Internet

2. Task

3. Process – Sequential

1. Summarizer Agent

LLM

Summarize

2. Task

Final
Outcome

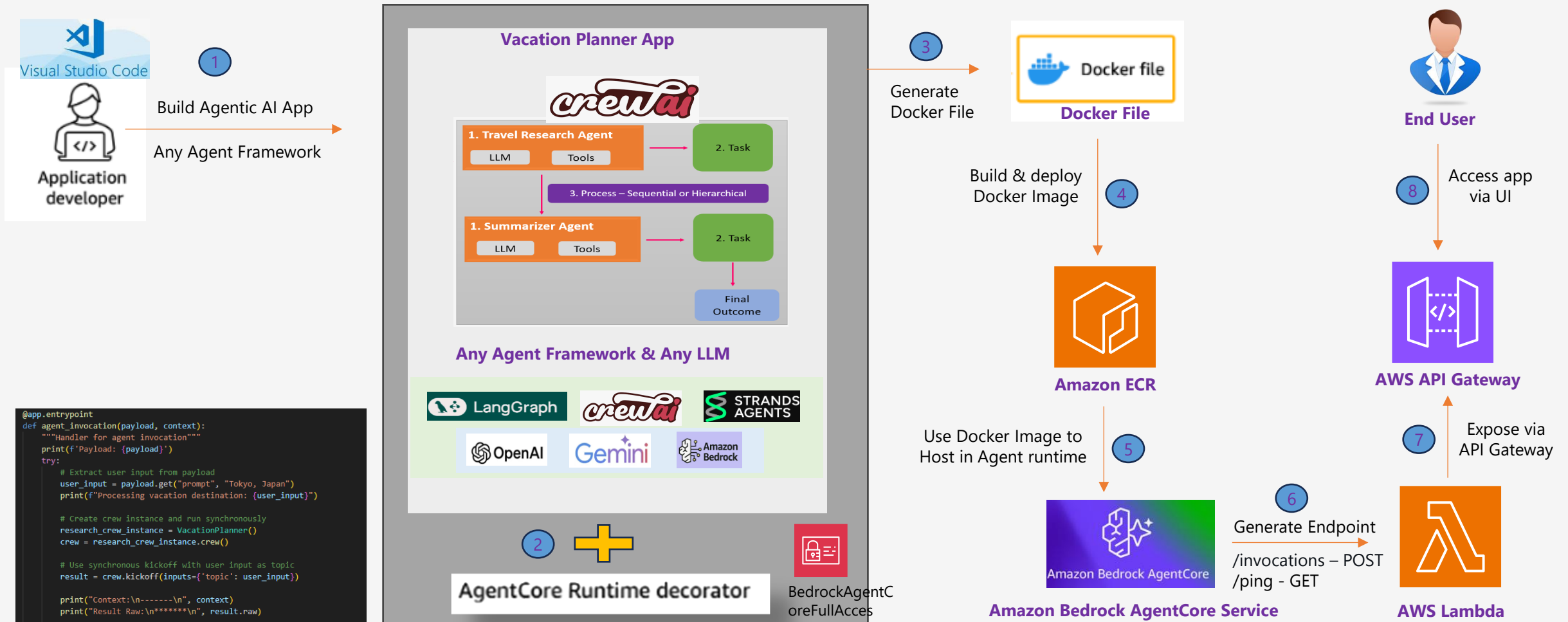




Section on :
Bedrock AgentCore Runtime

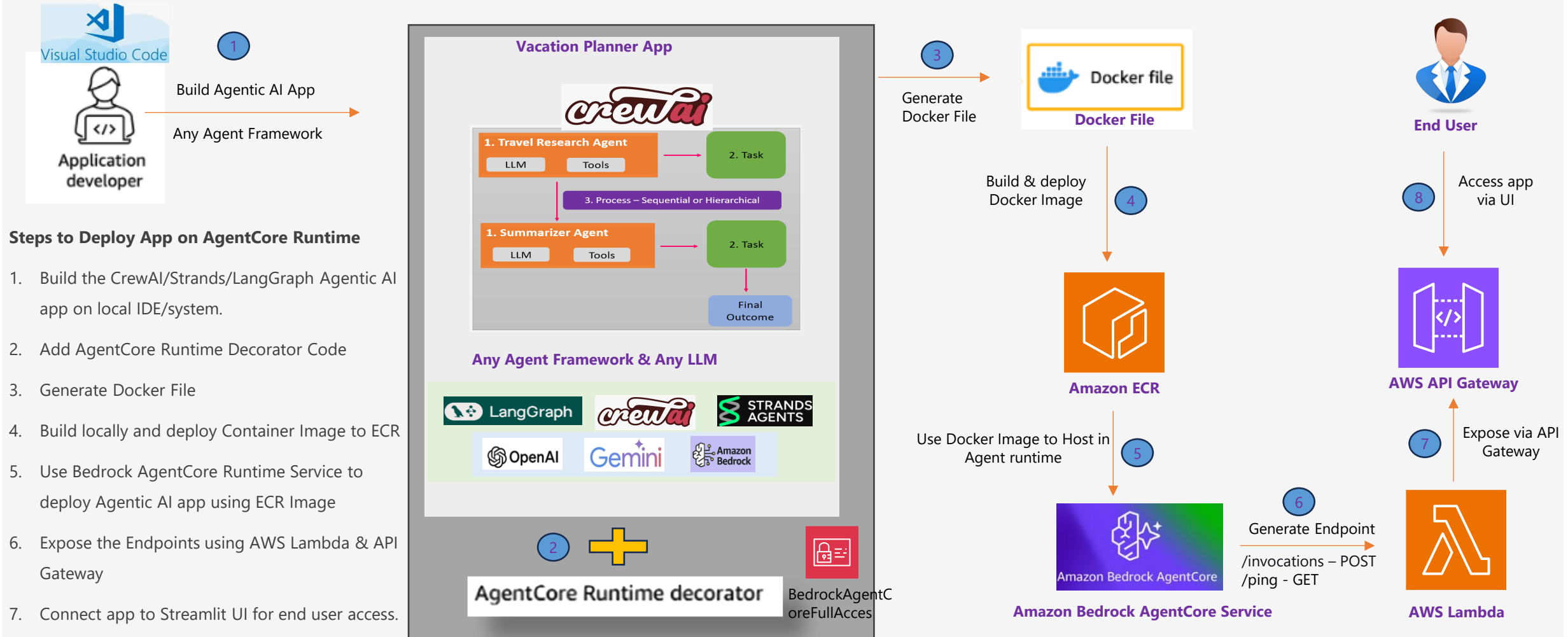
Bedrock AgentCore Capabilities – 1. AgentCore Runtime

Deploy Agentic AI App using Amazon Bedrock AgentCore Runtime



1. Bedrock AgentCore Runtime – Deployment Steps

Deploy Agentic AI App using Amazon Bedrock AgentCore Runtime





Section on :
Bedrock AgentCore Observability

Bedrock AgentCore – 2. Observability

- AgentCore Observability allows you to **trace, debug, and monitor AI agents' performance** in production environments.
- Agent Runtime, Memory, Gateways, Identity, Built-in tools
- **CloudWatch Logs – Agent & LLM (Model)**
- **CloudWatch Metrics**
- **GenAI Observability**
 - Model Invocations
 - Bedrock AgentCore

- Sessions - Represent complete user conversations
- Traces - Individual request-response cycles within a session
- Spans - Specific operations or steps within a trace

Session – Full interaction duration

Traces 1– Search Flight

Call API

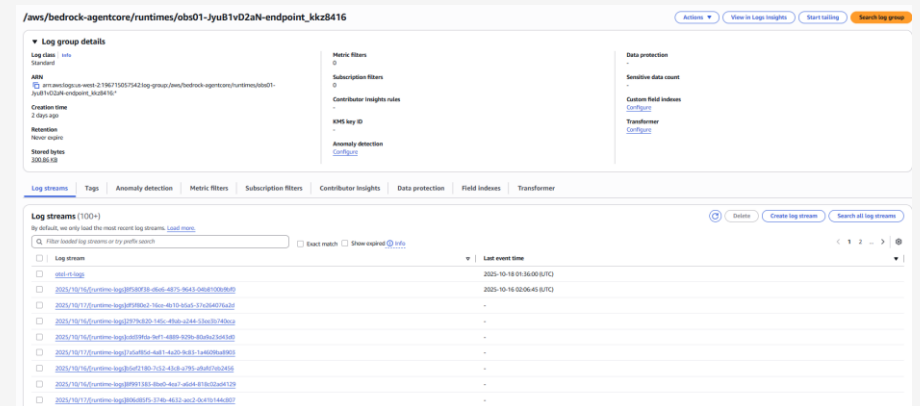
Return Result

Traces 2 – Book Flight

Pay

Book

CloudWatch Logs

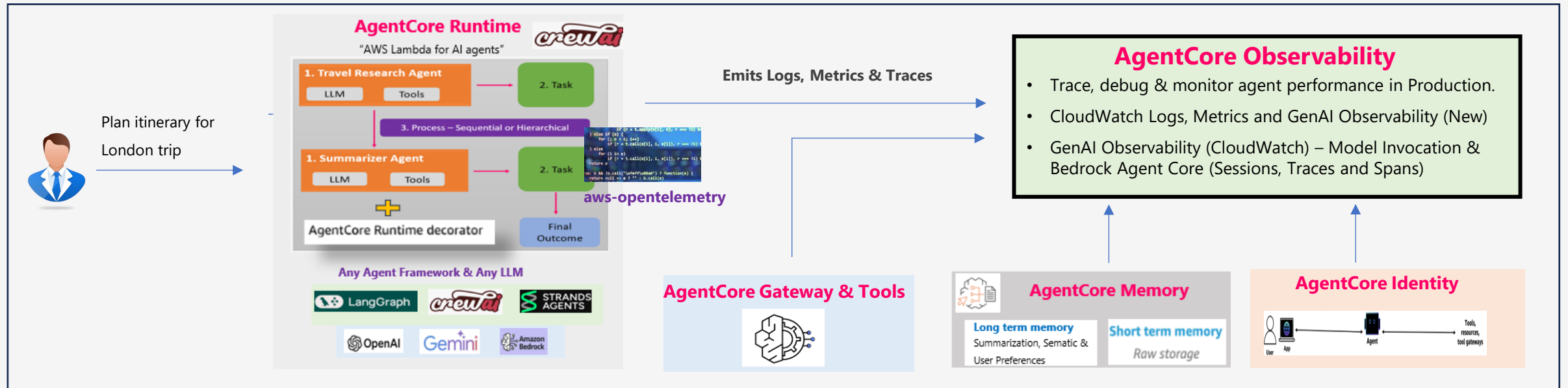


CloudWatch Metrics

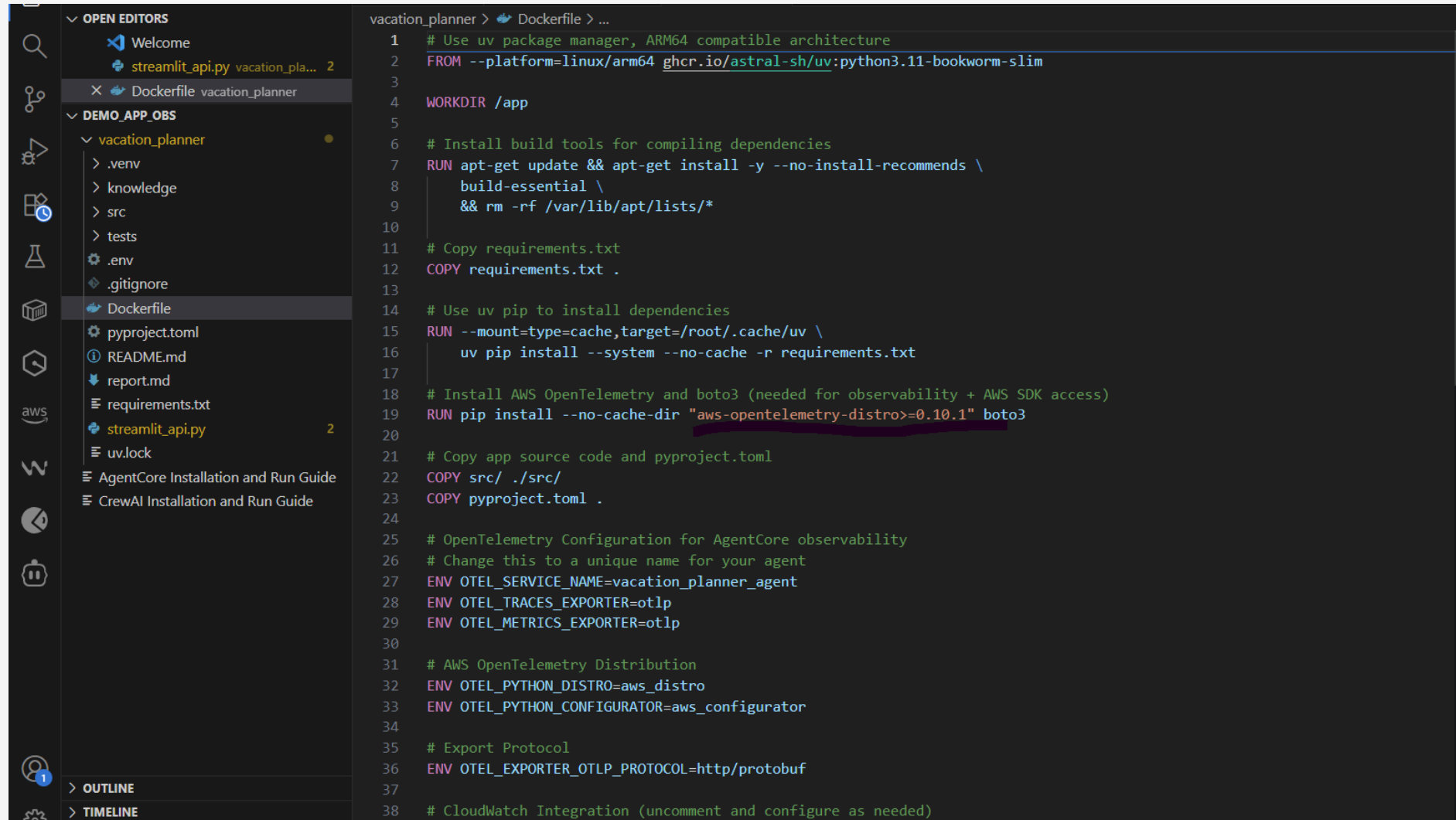
Browse (104)					
Multi source query					
Graphed metrics					
Options					
Source					
Oregon					
Search for any metric, dimension, resource id or account id					
Name	100/104	Operation	Resource	Metric name	Alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	UserErrors	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	SystemErrors	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Timeouts	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Durations	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Invocations	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Latency	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Sessions	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Errors	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Invocations	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	UserErrors	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Sessions	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Latency	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	SystemErrors	No alarms
agent0211111-DEFALC		InvokeAgentRuntime...	arn:aws:bedrock:agentcore-us-west-2:196735057542:runtime/agent0211111-1102QDvPh	Timeouts	No alarms

Bedrock AgentCore Capabilities – 2. Observability

- Amazon Bedrock AgentCore Observability enabled using **OpenTelemetry - Open-source observability framework**
- AWS Distro for OpenTelemetry (ADOT) - **aws-opentelemetry-distro >=0.10.1**
- Enabled for AgentCore Runtime, AgentCore Memory, AgentCore Identity and AgentCore Gateway



Bedrock AgentCore Capabilities – 2. Observability



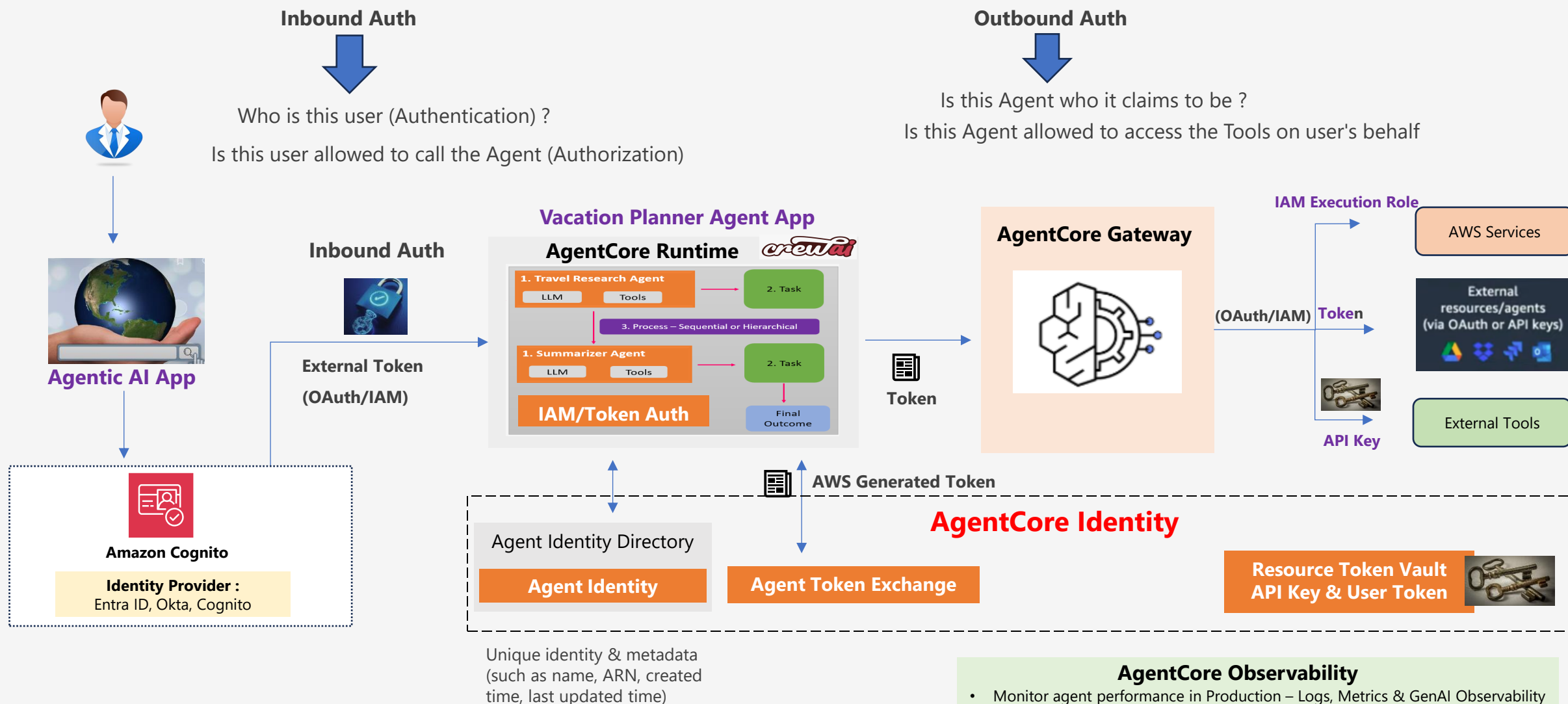
The screenshot shows a VS Code editor interface. On the left, the 'EXPLORER' sidebar displays a file tree for a project named 'vacation_planner'. The tree includes files like '.venv', 'knowledge', 'src', 'tests', '.env', '.gitignore', 'Dockerfile', 'pyproject.toml', 'README.md', 'report.md', 'requirements.txt', 'streamlit_api.py', and 'uv.lock'. The 'Dockerfile' is selected and open in the main editor. The editor shows the following Dockerfile content:

```
vacation_planner > Dockerfile > ...
1 # Use uv package manager, ARM64 compatible architecture
2 FROM --platform=linux/arm64 ghcr.io/astral-sh/uv:python3.11-bookworm-slim
3
4 WORKDIR /app
5
6 # Install build tools for compiling dependencies
7 RUN apt-get update && apt-get install -y --no-install-recommends \
8     build-essential \
9     && rm -rf /var/lib/apt/lists/*
10
11 # Copy requirements.txt
12 COPY requirements.txt .
13
14 # Use uv pip to install dependencies
15 RUN --mount-type=cache,target=/root/.cache/uv \
16     uv pip install --system --no-cache -r requirements.txt
17
18 # Install AWS OpenTelemetry and boto3 (needed for observability + AWS SDK access)
19 RUN pip install --no-cache-dir "aws-opentelemetry-distro>=0.10.1" boto3
20
21 # Copy app source code and pyproject.toml
22 COPY src/ ./src/
23 COPY pyproject.toml .
24
25 # OpenTelemetry Configuration for AgentCore observability
26 # Change this to a unique name for your agent
27 ENV OTEL_SERVICE_NAME=vacation_planner_agent
28 ENV OTEL_TRACES_EXPORTER=otlp
29 ENV OTEL_METRICS_EXPORTER=otlp
30
31 # AWS OpenTelemetry Distribution
32 ENV OTEL_PYTHON_DISTRO=aws_distro
33 ENV OTEL_PYTHON_CONFIGURATOR=aws_configurator
34
35 # Export Protocol
36 ENV OTEL_EXPORTER_OTLP_PROTOCOL=http/protobuf
37
38 # CloudWatch Integration (uncomment and configure as needed)
```



Section on :
Bedrock AgentCore Identity

Bedrock AgentCore – Identity (Inbound and Outbound)



Bedrock AgentCore – Identity (Inbound and Outbound)

- **Amazon Bedrock AgentCore Identity** is an agent identity and access management capability that helps you securely build enterprise-ready agents

Key Features

- **Inbound Authentication** : Validate access for users and applications calling agents or tools
- **Outbound Authentication** : Secure access from agents to external services on behalf of users
- **Supports OAuth Integration** : Enterprise ready
- **AWS IAM Integration** : Native integration with AWS identity and access management
- **Cross-Platform Support** : Works across AWS, other cloud providers, and on-premise environments

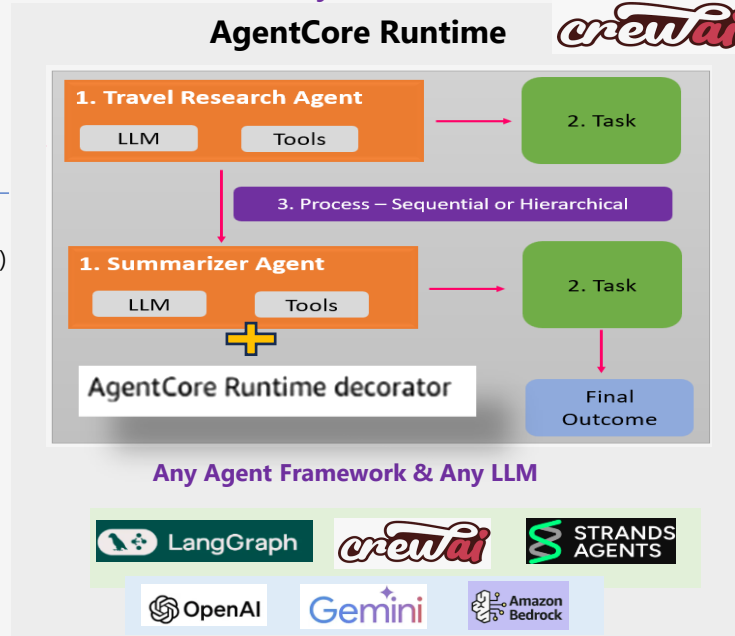
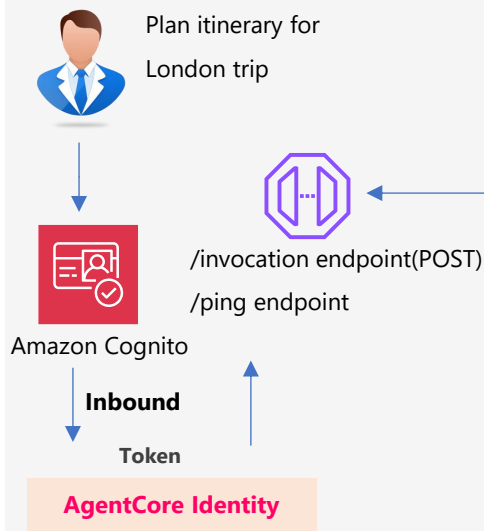




Section on :
Bedrock AgentCore Gateway

Bedrock AgentCore – AgentCore Gateway

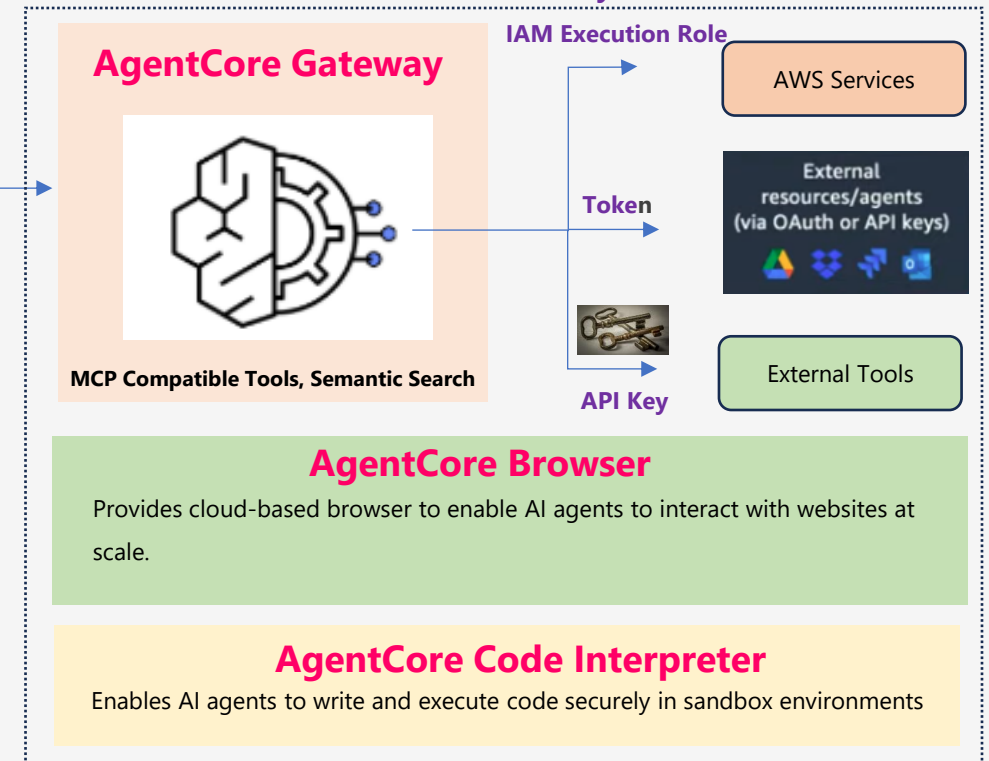
Inbound – Access to Gateway



Access Tools
(MCP Protocol & Discovery)

Outbound

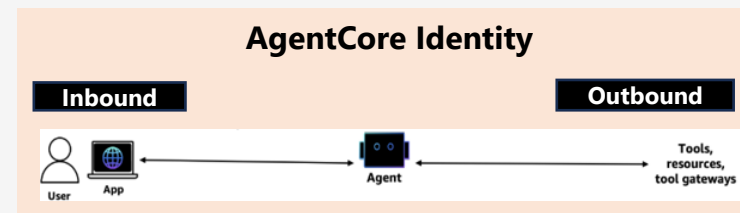
Outbound – Allow/Deny Access to resources



Integration

AgentCore Gateway Capabilities

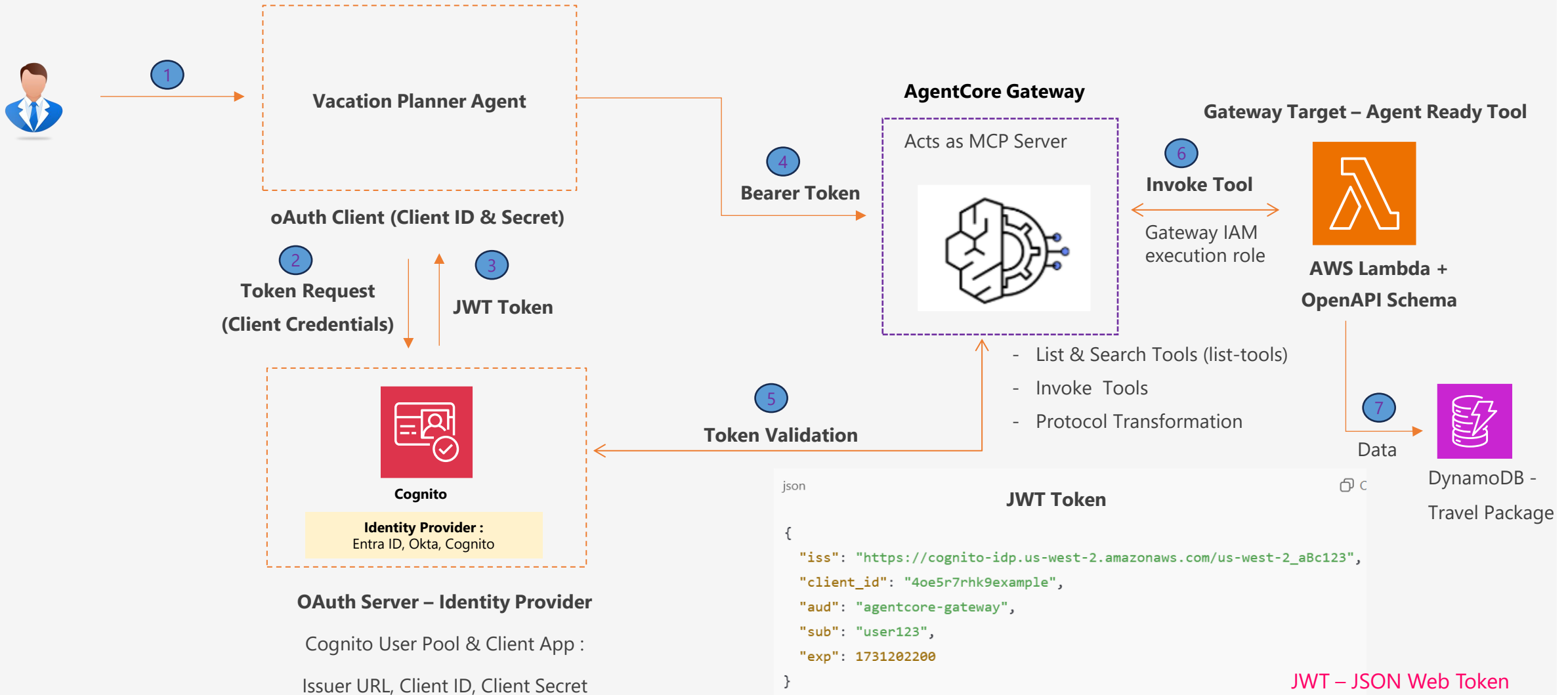
1. Security – User – Agent- Tool Access (IAM & OAuth)
2. Intelligent Tool Discovery – Semantic Search (Vector Embeddings)
3. MCPify Tools (Protocol Conversion)
4. Observability
5. Serverless Infrastructure Management



AgentCore Observability

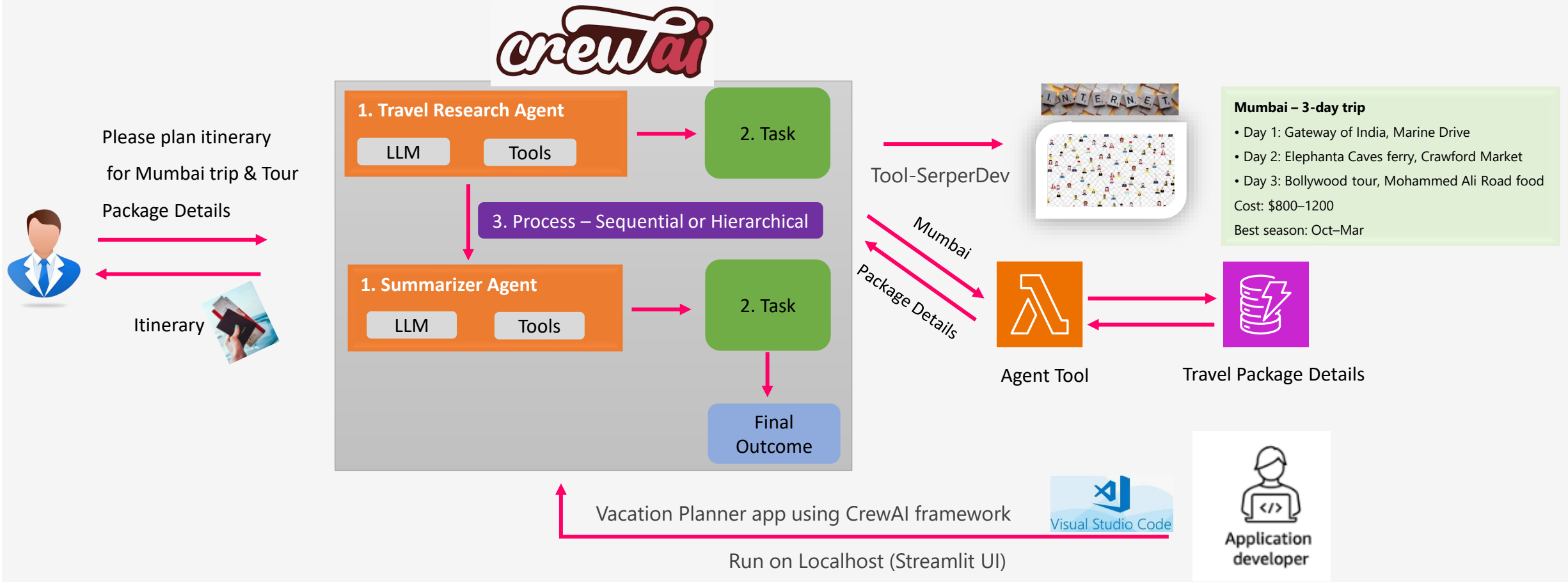
Monitor agent performance in Production – Logs, Metrics & GenAI Observability

Bedrock AgentCore Identity and Gateway – Flow



Bedrock AgentCore – Business Scenario : PoC to Production

Business Use Case : Dream Vacation Planner travel agency wants to build and test (PoC) an Agentic AI app which allows its users to plan an itinerary for their dream destination on their website. *Get city travel package information including travel plan, cost, duration and best season.*





Section on :
Bedrock AgentCore Memory

Bedrock AgentCore Capabilities – Short Term & Long-Term Memory

Challenges with LLM :

- **LLM's are stateless** - Each interaction as a new instance with no knowledge of previous conversations
- No Short term or Long-term memory
- Developers need to build custom memory

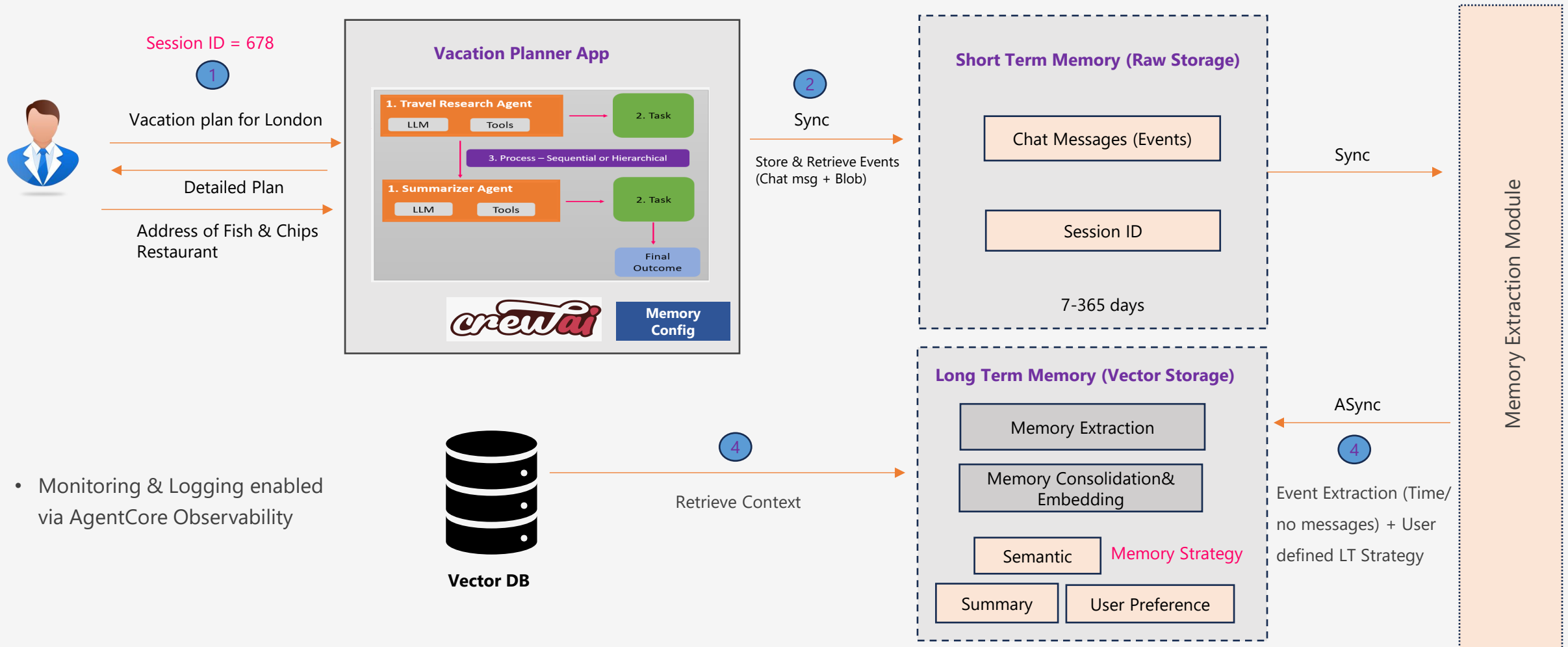


Bedrock AgentCore Memory

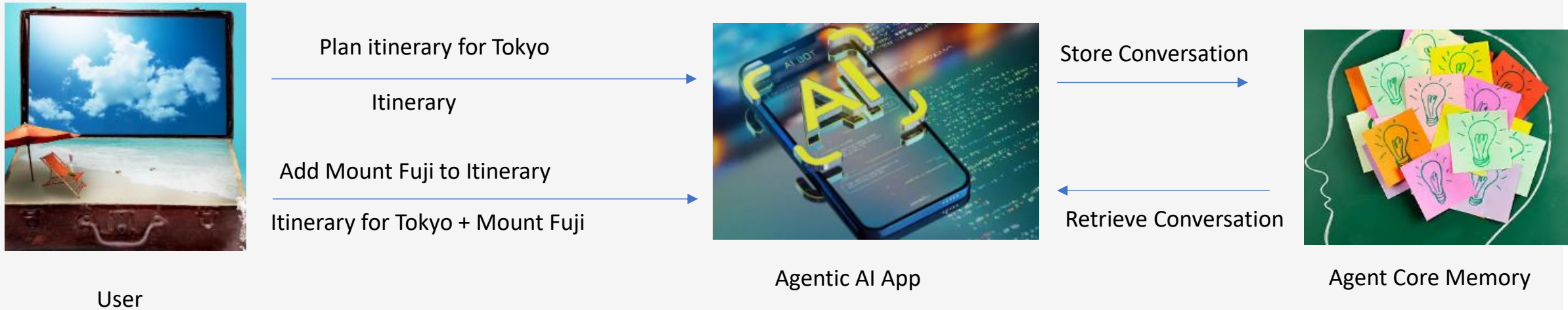
- **Fully managed service** that gives your AI agents the ability to remember past interactions
- Short-Term and Long-Term Memory



Bedrock AgentCore Capabilities – Short Term & Long-Term Memory



Bedrock AgentCore Capabilities – Short Term & Long-Term Memory



Bedrock AgentCore Capabilities – Short Term & Long-Term Memory



User

Actor Id= User

Session ID = 24526

Event ID = 123

Plan itinerary for Tokyo

Itinerary

Event ID = 234

Add Mount Fuji to Itinerary

Itinerary for Tokyo + Mount Fuji



Agentic AI App

Agent ID = Vacation Planner

Store Conversation

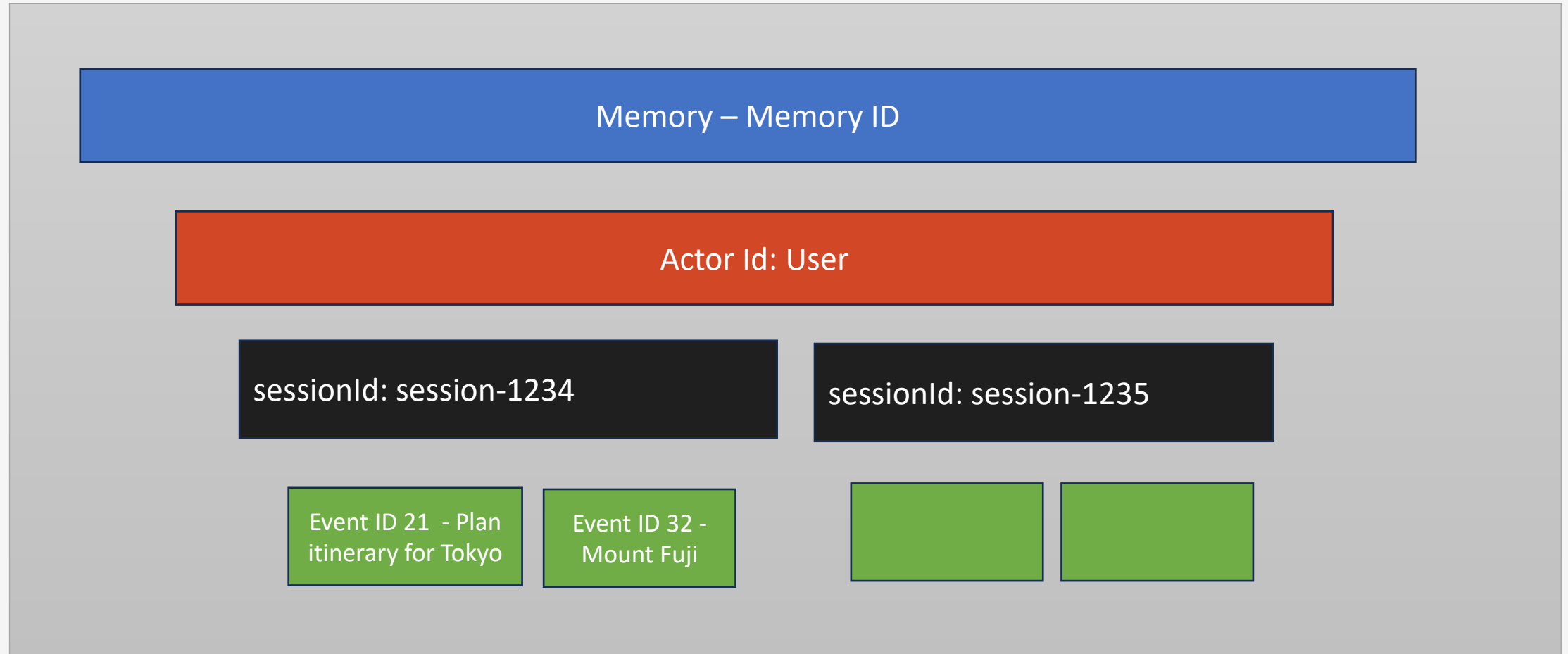
Retrieve Conversation



Agent Core Memory

Memory ID = abc24526

Bedrock AgentCore Capabilities – Short Term & Long-Term Memory



Thank You